IV Bolus Practice 4

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

1a 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. a. What is the expected initial concentration of the drug in the plasma?

1b 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. b. What percent of the dose is eliminated in 24 hours?

1c 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. c. What is the expected plasma drug concentration (C_p) at 12 hours after administration of the dose?

1d 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. d. If the dose were doubled, what is the expected change in the half-life of elimination?

A. The half-life increases, but by less than double

B. No change; the half-life is constant for a first-order process

C. The half-life doubles

D. The half-life halves

1e 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. e. If the dose were doubled, what is the expected change in the clearance?

A. Clearance halves

B. No change; clearance is constant for a first-order process

C. Clearance doubles

D. Clearance increases by a factor of 4

1f 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. f. If the dose were doubled, what is the expected change in the initial plasma concentration?

A. No change

B. It increases by a factor of 4

C. It halves

D. It increases by a factor of 2

2a 2. The pharmacokinetics of an antibiotic administered by IV bolus injection is best described by the equation below. (Concentration is given in mcg/mL and time in hours)

$C_p = 18.35e^{-4.8t} + 12.63e^{-0.09t}$

a. What is the elimination half-life of this drug?

2b 2. The pharmacokinetics of an antibiotic administered by IV bolus injection is best described by the equation below. (Concentration is given in mcg/mL and time in hours)

$C_p = 18.35e^{-4.8t} + 12.63e^{-0.09t}$

b. What is the initial plasma concentration of this drug?

2c 2. The pharmacokinetics of an antibiotic administered by IV bolus injection is best described by the equation below. (Concentration is given in mcg/mL and time in hours)

$C_p = 18.35e^{-4.8t} + 12.63e^{-0.09t}$

c. What is the concentration of drug in the plasma 4 hours after administration of the dose?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a Her key

Answer: 24.19 mg/L

WHAT THE STEM GIVES

t½	6 hours	*not needed for C_0*
V_D	400 mL/kg	0.4 L/kg × 62 kg = 24.8 L, the denominator
D_0	600-mg	the numerator
weight	62 kg	scales the volume

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = (400\ \text{mL/kg})\left(\frac{1\ \text{L}}{1000\ \text{mL}}\right)(62\ \text{kg}) = (0.4\ \text{L/kg})(62\ \text{kg}) = 24.8\ \text{L}$$

mL/kg to L/kg divides by 1000; multiplying by the 62 kg weight cancels the kilograms and leaves litres.

SETUP

$$C_0 = \frac{D_0}{V_D}$$

Right after a rapid IV injection the whole dose is in the volume of distribution, so the concentration at time zero is dose over volume.

ALGEBRA

$$C_0 = \frac{600\ \text{mg}}{24.8\ \text{L}} = 24.19\ \text{mg/L}$$

mg over L leaves mg/L: 24.19 mg of drug in each litre of the 24.8 L volume.

> **DOES THE NUMBER MAKE SENSE?**
> Each litre of the 24.8 L holds a little over 24 mg, because 24.8 × 24 is just under 600 mg; 24.19 mg/L fits.

1b Her key

Answer: 93.75 %

WHAT THE STEM GIVES

t½	6 hours	24 hours is four half-lives; k = 0.1155 hr^{-1}
V_D	400 mL/kg	*not needed: a percent of the dose*
D_0	600-mg	*not needed: the percent is the same for any dose*
weight	62 kg	*not needed*
t	24 hours	four half-lives

THE WORKING, ONE LINE AT A TIME

SETUP

$24 \text{ hr} \div 6 \text{ hr} = 4$ half-lives

The half-life is 6 hours, so 24 hours holds four of them; the hours cancel and a count is left.

SETUP

$\%\text{ remaining} = 100\ e^{-kt}$, with $k = \dfrac{0.693}{6 \text{ hr}} = 0.1155 \text{ hr}^{-1}$

First-order loss: the fraction of the dose left after t is e^{-kt}, so $100\ e^{-kt}$ is the percent remaining.

ALGEBRA

$100\ e^{-(0.1155)(24)} = 100\ e^{-2.772} = 6.25\%$ remaining

$kt = (0.1155 \text{ hr}^{-1})(24 \text{ hr}) = 2.772$ has no units. Four halvings leave one sixteenth of the dose, 6.25%.

ALGEBRA

$100\% - 6.25\% = 93.75\%$ eliminated

What has been eliminated is what no longer remains: 93.75% of the dose in 24 hours.

> **DOES THE NUMBER MAKE SENSE?**
> Each 6-hour half-life removes half of what is left: 50%, 75%, 87.5%, then 93.75% after the fourth, at 24 hours.

Her key prints the last line as "100% − 9.25%"; her line above gives 6.25% remaining, and 100 − 6.25 = 93.75, her answer.

1c Her key

Answer: 6.05 mg/L

WHAT THE STEM GIVES

$t_{½}$	6 hours	$k = 0.1155 \text{ hr}^{-1}$; 12 hours is two half-lives
V_D	400 mL/kg	gave C_0 in part a
D_0	600-mg	gave C_0 in part a
weight	62 kg	gave $V_D = 24.8$ L in part a
t	12 hours	the exponent kt
C_0	24.19 mg/L	multiplied by e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_0\ e^{-kt}$, with $k = \dfrac{0.693}{6 \text{ hr}} = 0.1155 \text{ hr}^{-1}$

After an IV bolus the concentration declines first order from C_0, so C_0 is multiplied by the fraction left at time t.

ALGEBRA

$C12 = (24.19 \text{ mg/L})\ e^{-(0.1155)(12)} = (24.19 \text{ mg/L})(0.25) = 6.05 \text{ mg/L}$

$kt = (0.1155 \text{ hr}^{-1})(12 \text{ hr}) = 1.386$, a pure number; $e^{-1.386} = 0.25$, so the answer keeps the mg/L of C_0.

ALGEBRA

12 hr is 2 half-lives: 25% remains, (0.25)(24.19 mg/L) = 6.05 mg/L

Her key also counts half-lives: two halvings leave a quarter of C_0.

> **DOES THE NUMBER MAKE SENSE?**
> Twelve hours is two half-lives of 6 hours, so a quarter of 24.19 mg/L remains, a little over 6 mg/L; 6.05 mg/L fits.

1d

Her key

Answer: B. No change; the half-life is constant for a first-order process

WHAT THE STEM GIVES

t½	6 hours	unchanged: $\frac{0.693}{k}$ has no dose in it
D_0	600-mg	doubled; it does not enter k or t½
V_D	400 mL/kg	*not needed: the volume does not enter t½*
weight	62 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{½} = \frac{0.693}{k} = \frac{0.693}{0.1155\ hr^{-1}} = 6\ hr \text{ for either dose}$$

The half-life depends only on k, and the dose does not appear in it, so 600 mg and 1200 mg have the same 6-hour half-life.

> **DOES THE NUMBER MAKE SENSE?**
> For a first-order drug k, t½, V_D and clearance are constants of the drug in the patient; only amounts and concentrations scale with the dose.

No change in half-life: half-life is constant for first-order processes. $t_{½} = \frac{0.693}{k}$, and k is a property of the drug in this patient, not of the dose.

1e

Her key

Answer: B. No change; clearance is constant for a first-order process

WHAT THE STEM GIVES

t½	6 hours	$k = 0.1155\ hr^{-1}$, a constant
V_D	400 mL/kg	24.8 L for 62 kg, a constant
D_0	600-mg	doubled; it does not enter Cl_T
weight	62 kg	times 0.4 L/kg gives the 24.8 L in Cl_T

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$Cl_T = k \times V_D = (0.1155\ hr^{-1})(24.8\ L) = 2.86\ L/hr$ for either dose

Neither factor contains the dose: hr^{-1} times L gives L/hr, the same volume of plasma cleared each hour for 600 mg or 1200 mg.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance is a volume per hour, $k \times V_D$, and neither factor changes with the dose; the mg removed per hour, $Cl_T \times C_p$, is what doubles.

No change in clearance: clearance is constant for first-order processes. $Cl_T = k \times V_D$, and neither k nor V_D depends on the dose.

1f

Her key

Answer: D. It increases by a factor of 2

WHAT THE STEM GIVES

V_D	400 mL/kg	24.8 L, unchanged
D_0	600-mg	doubled, the numerator of the new C_0
C_0	24.19 mg/L	the value that doubles
t½	6 hours	*not needed: C_0 comes before any elimination*
weight	62 kg	times 0.4 L/kg gives V_D = 24.8 L

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{D_0}{V_D}, \text{ with } V_D = (0.4\ L/kg)(62\ kg) = 24.8\ L \text{ unchanged}$$

The initial concentration is dose over volume; the volume is a property of the drug in the patient and stays at 24.8 L.

ALGEBRA

$$C_0 = \frac{1200\ mg}{24.8\ L} = 48.387\ mg/L$$

mg over L gives mg/L. 48.387 mg/L is twice the 24.19 mg/L of part a.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 is proportional to the dose when V_D is fixed, so the new C_0, 48.387 mg/L, is twice the 24.19 mg/L of part a.

Expect the initial concentration to increase by a factor of 2. $C_0 = \frac{D_0}{V_D}$ with V_D unchanged, so 1200 mg gives 48.387 mg/L, twice 24.19 mg/L.

2a

Her key

Answer: 7.7 hr

WHAT THE STEM GIVES

A	18.35	*not needed*
α	4.8	*not needed: the distribution exponent*
B	12.63	*not needed*
β	0.09	the denominator of $\frac{0.693}{\beta}$
units	mcg/mL and hours	β is in hr^{-1}; 1 mcg/mL = 1 mg/L

THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = A\,e^{-\alpha t} + B\,e^{-\beta t}$: $\alpha = 4.8\ hr^{-1}$, $\beta = 0.09\ hr^{-1}$

The two exponents are read from the equation; the smaller, 0.09 hr^{-1}, is β, the elimination rate constant of the terminal phase.

ALGEBRA

$$t½\beta = \frac{0.693}{0.09\ hr^{-1}} = 7.7\ hr$$

A pure number over hr^{-1} leaves hours. $\alpha = 4.8\ hr^{-1}$ describes distribution and is not used.

> **DOES THE NUMBER MAKE SENSE?**
> $\beta = 0.09\ hr^{-1}$ is a slow rate, so the half-life is several hours: 0.693 over 0.09 is 7.7 hr. A half-life from $\alpha = 4.8\ hr^{-1}$ would be minutes.

Her sheet prints the second term as $12.63e^{-0.09}$, without t; her part c evaluates $e^{-0.09}(4)$, so the term is $12.63e^{-0.09t}$.

2b Her key

Answer: 30.98 mcg/mL

WHAT THE STEM GIVES

A	18.35	added
α	4.8	*not needed: $e^0 = 1$*
B	12.63	added
β	0.09	*not needed: $e^0 = 1$*
units	mcg/mL and hours	β is in hr^{-1}; 1 mcg/mL = 1 mg/L

THE WORKING, ONE LINE AT A TIME

SETUP

$C_0 = A + B$

At t = 0 both exponentials equal 1, so the initial concentration is the sum of the two coefficients.

ALGEBRA

$C_0 = (18.35 + 12.63)\ mcg/mL = 30.98\ mcg/mL$

Both coefficients are in mcg/mL, so they add. Her key labels the result 30.98 mg/L, the same value, because 1 mcg/mL = 1 mg/L.

> **DOES THE NUMBER MAKE SENSE?**

> C_0 is the sum of the two coefficients, so it exceeds each: 18.35 + 12.63 = 30.98 mcg/mL.

2c Her key

Answer: 8.812 mcg/mL

WHAT THE STEM GIVES

A	18.35	the distribution term, negligible by 4 hours
α	4.8	αt = 19.2
B	12.63	the elimination term, almost all of C_p at 4 hours
β	0.09	βt = 0.36
units	mcg/mL and hours	β is in hr^{-1}; 1 mcg/mL = 1 mg/L
t	4 hours	the time in both exponents

THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = A\,e^{-\alpha t} + B\,e^{-\beta t}$, at t = 4 hr

The two-compartment equation is evaluated at 4 hours; each exponent is hr^{-1} times hr, a pure number.

ALGEBRA

$C4 = (18.35\ mcg/mL)\,e^{-(4.8)(4)} + (12.63\ mcg/mL)\,e^{-(0.09)(4)} = (18.35\ mcg/mL)\,e^{-19.2} + (12.63\ mcg/mL)\,e^{-0.36}$

The distribution exponent is 19.2 and the elimination exponent 0.36.

ALGEBRA

$C4 = 0.0000001\ mcg/mL + (12.63\ mcg/mL)(0.6977) = 8.812\ mcg/mL$

$e^{-19.2}$ is negligible, so by 4 hours only the β term remains. Her key labels the 8.812 in mg/L, the same value as mcg/mL.

> **DOES THE NUMBER MAKE SENSE?**
>
> By 4 hours the 4.8 hr^{-1} term is negligible, so C_p is about 12.63 times $e^{-0.36}$, a little over two thirds of 12.63: 8.812 mcg/mL.

Her sheet prints the second term as $12.63e^{-0.09}$, without t; her part c evaluates $e^{-0.09}(4)$, so the term is $12.63e^{-0.09t}$.
